Fused Sampler

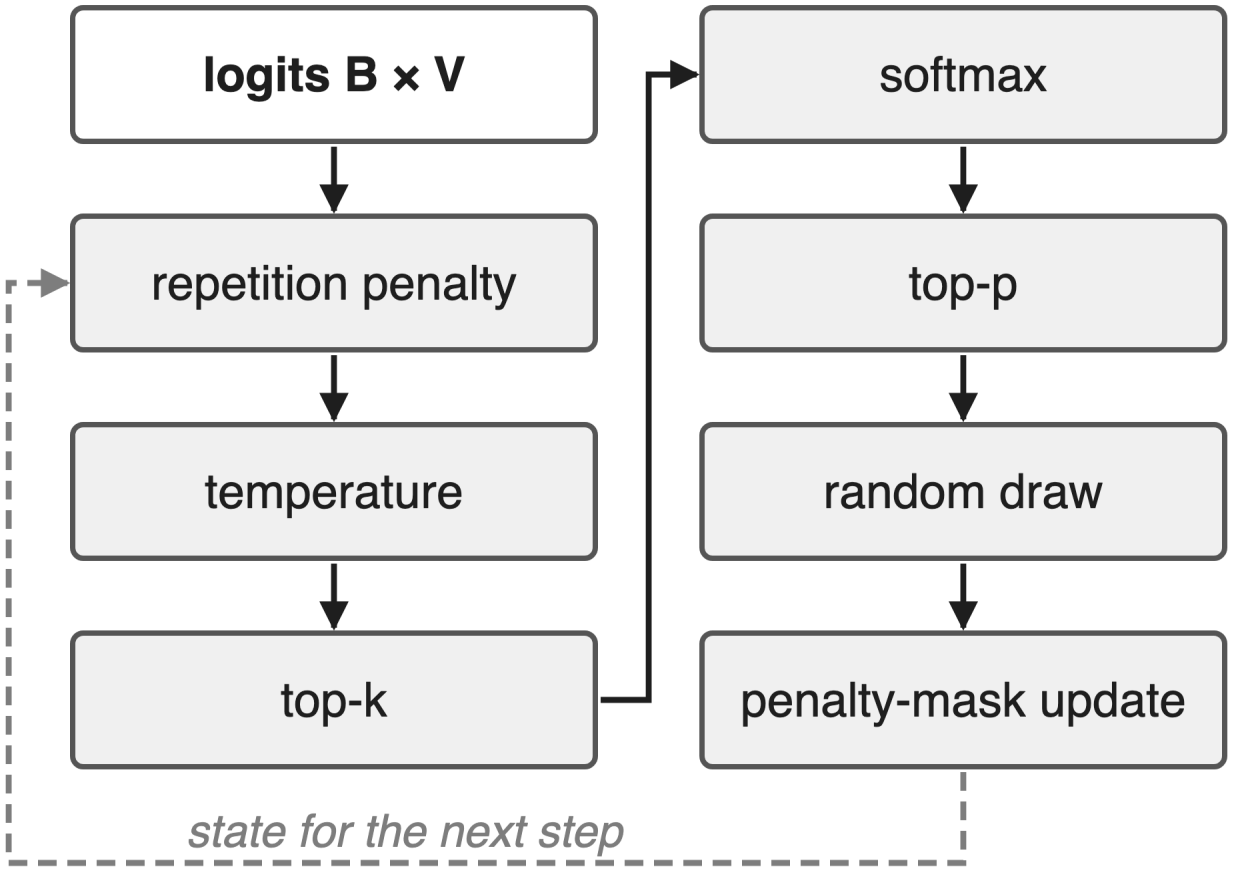
Decode-step sampling in two CUDA kernels

Tencent Hunyuan AI Infra · HPC-Ops

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Every decode step ends in a sampler

- B × V logits in, B tokens out; V = **120,832** here
- Seven stages, once per generated token
- Small batches expose fixed launch and pass costs



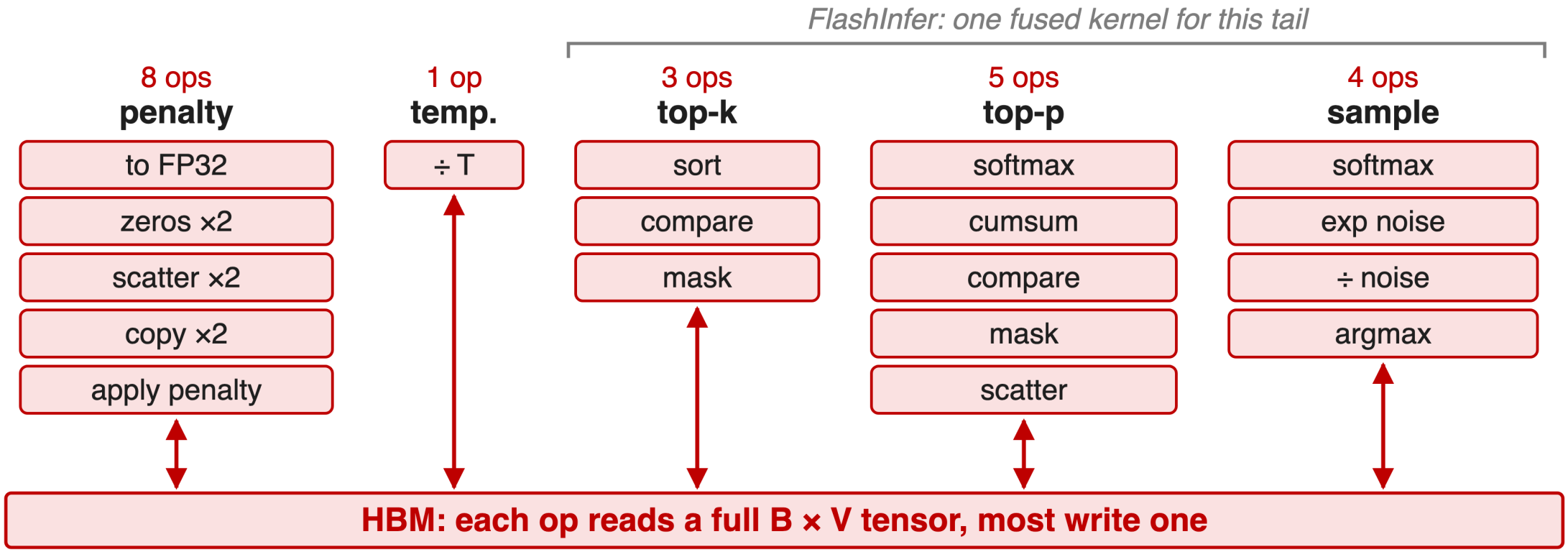
A PyTorch sampler runs ~20 vocabulary-wide ops

~20

separate tensor ops per decode step in a vLLM-style PyTorch sampler, each over a B × 120,832 tensor

Counted from HPC-Ops benchmark/sampler/benchmark_sampler.py (full sampling: repetition penalty, temperature, top-k 20, top-p 0.9)

Each stage re-reads the full vocabulary



Top-k keeps 20 tokens, yet every op touches all 120,832

Insight: after top-k, at most 64 tokens matter

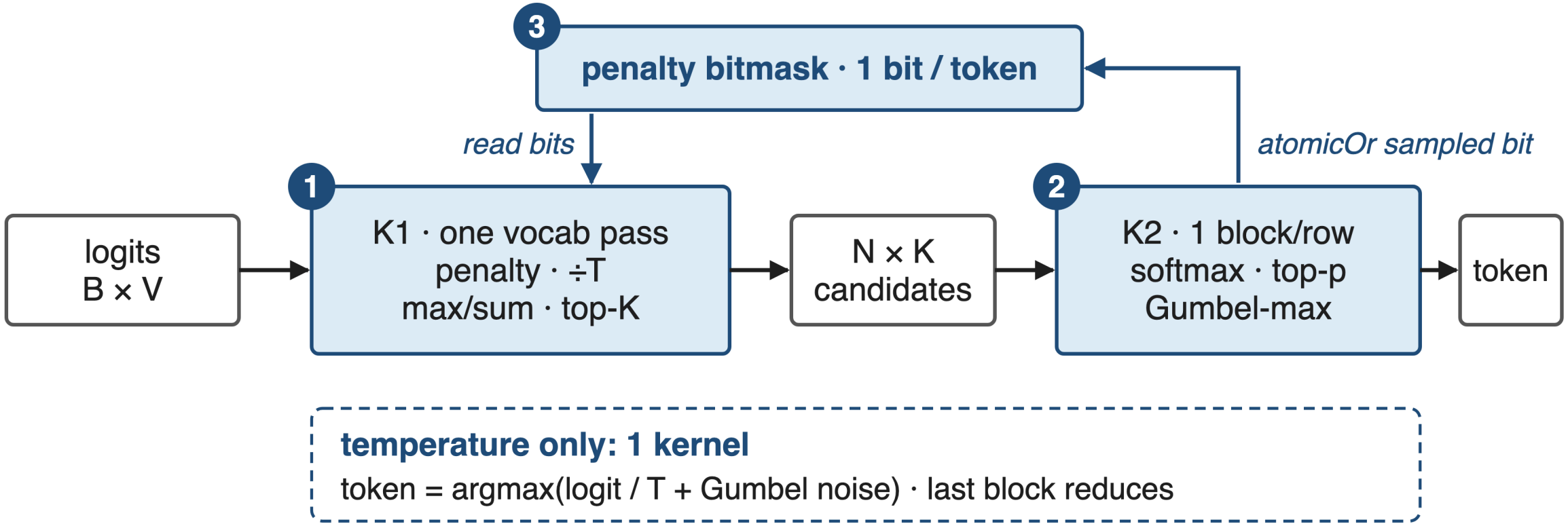
- Penalty, temperature: elementwise, fold into the read
- Softmax normalizer: online max + sum
- Sampling: argmax(log p + Gumbel noise)

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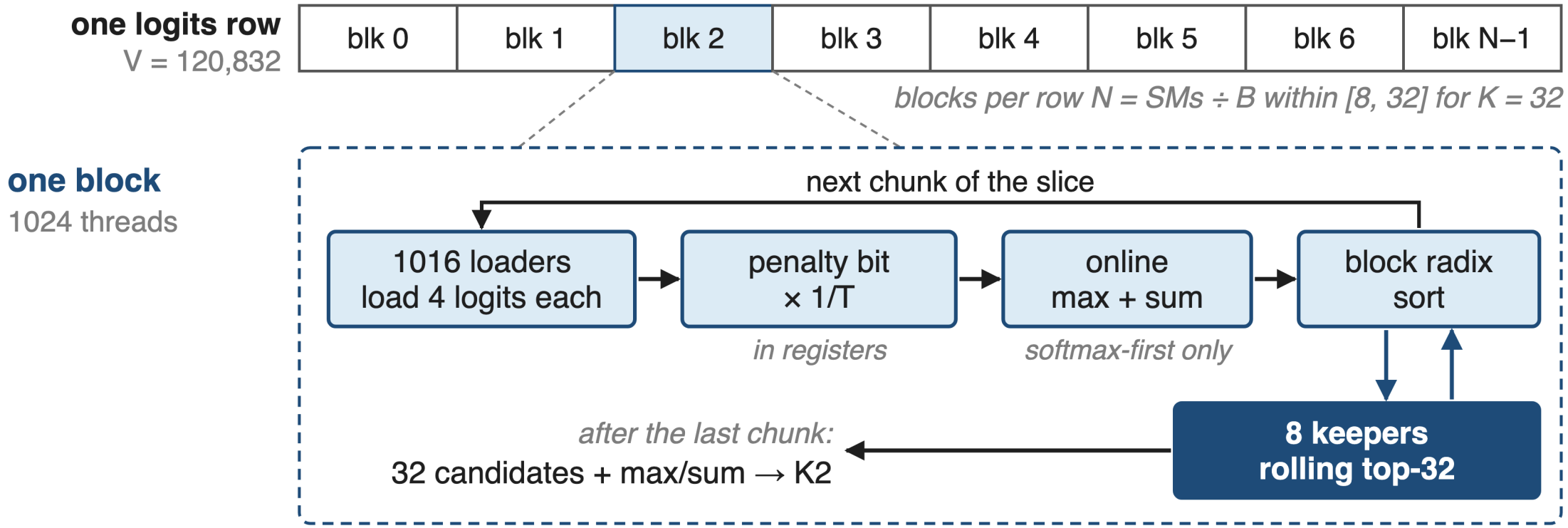
One read of the logits, then work on ≤ 64 candidates per row

Two kernels: one vocabulary pass, one tiny tail



API: hpc.fused_sampler(logits, ...) · kernels: src/sampler/fused_sampler.cu and fused_sampler_temperature.cu

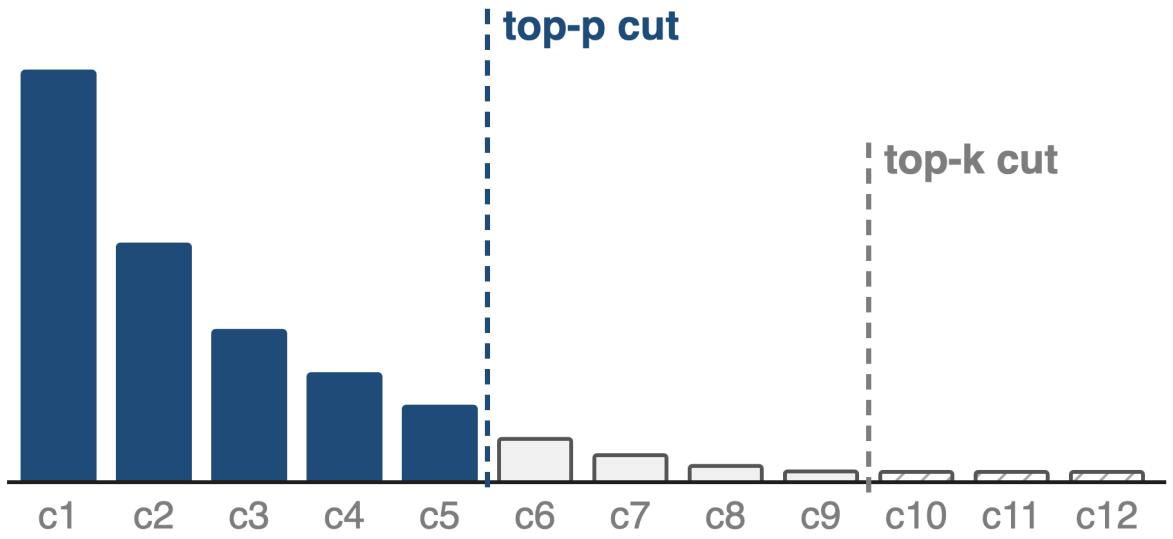
❶ K1 reads each logit exactly once



A block sort per chunk keeps the running top-K; V is never fully sorted

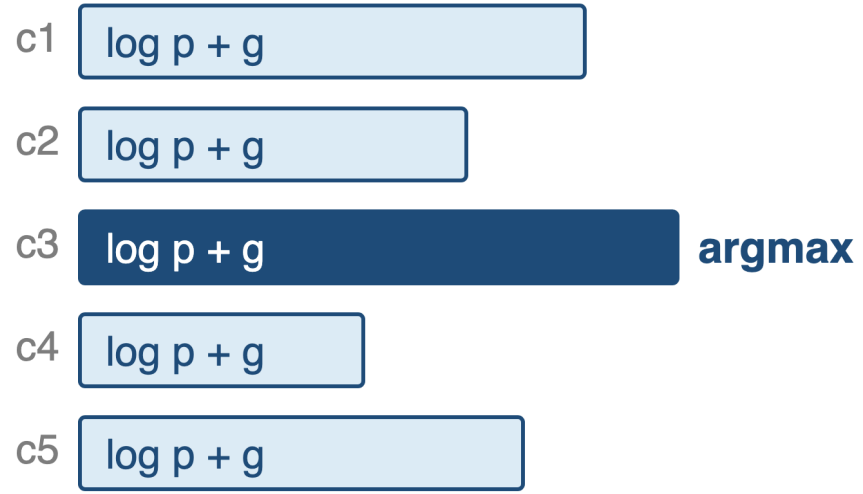
❷ K2 samples by Gumbel-max over survivors

sorted candidates of one row



softmax over the top-k · prefix sum by warp scan

Gumbel-max over survivors

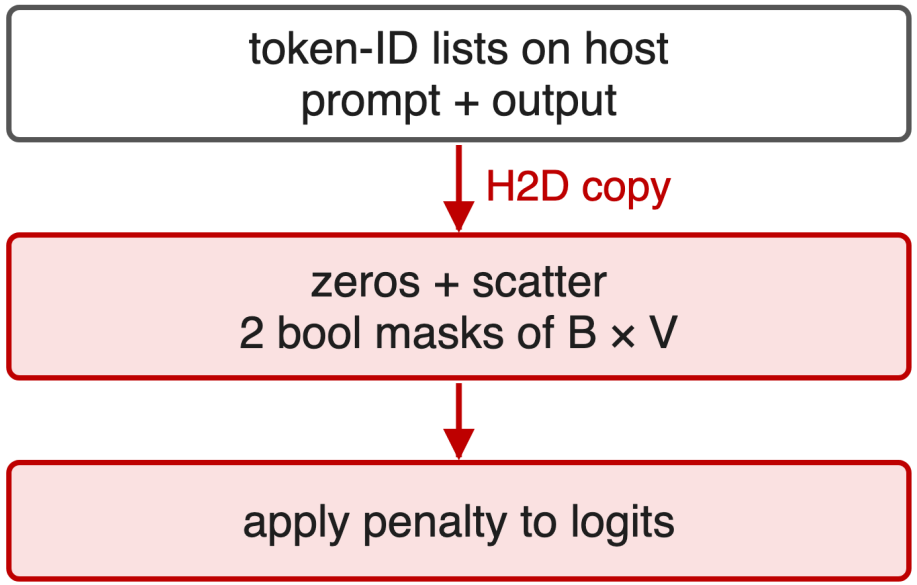


g = Gumbel noise per candidate

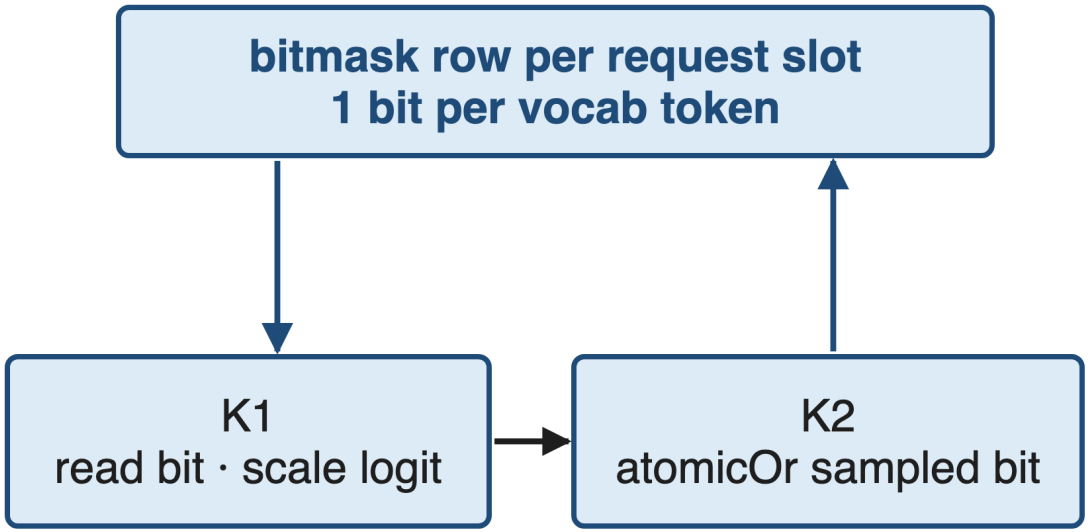
argmax(log p_i + g_i) picks token i with probability p_i

❸ The penalty mask never leaves the GPU

vLLM-style: rebuilt every step



HPC-Ops: resident 1-bit mask

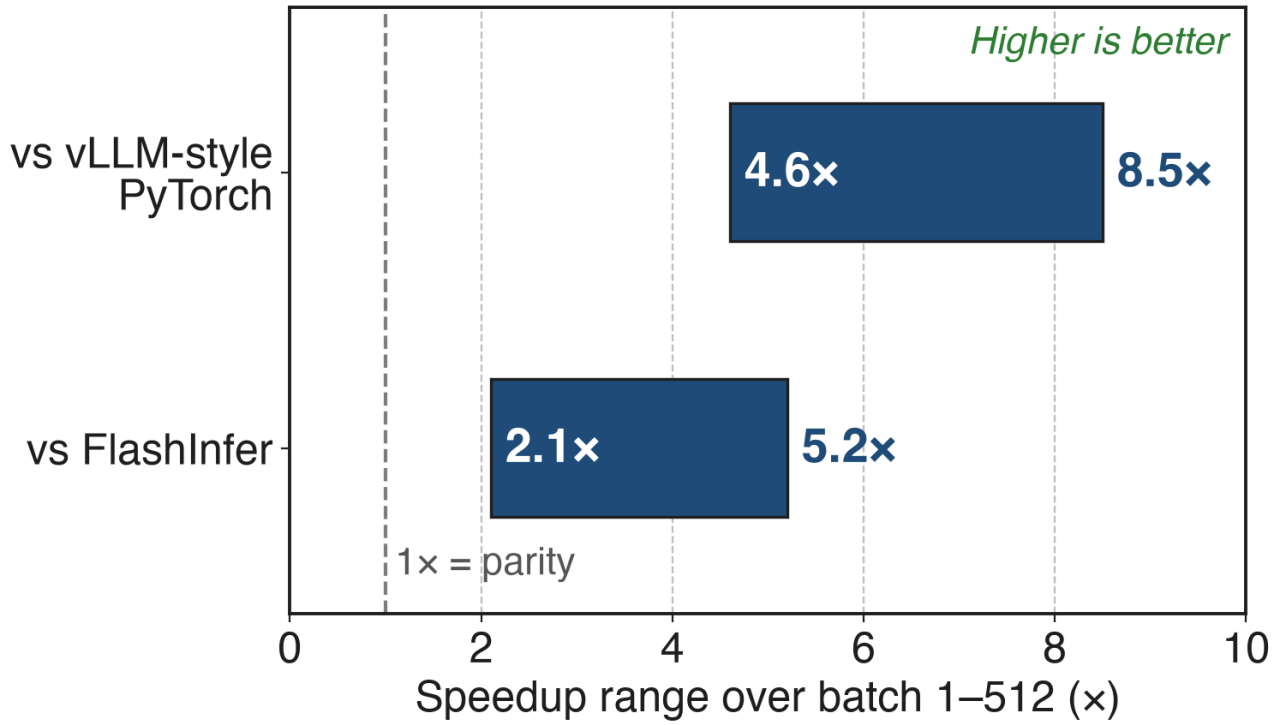


row = V / 8 = 15,104 bytes · no host copy

One bit per token: K2 sets it, K1 reads it on the next step

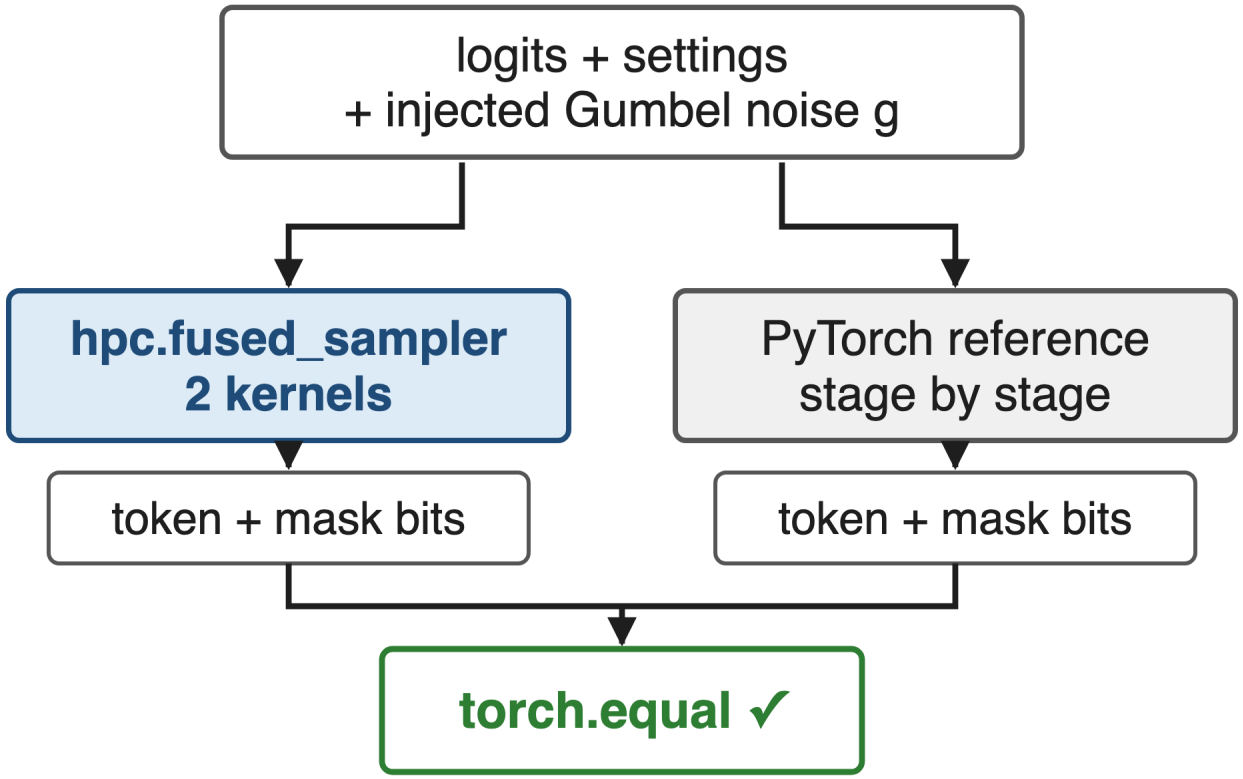
Runs 4.6–8.5× faster than a vLLM-style sampler

- BF16 logits, V = 120,832, batch 1–512
- Penalty 1.1, T 1.05, top-k 20, top-p 0.9
- **2.1–5.2×** faster than the FlashInfer path



Same noise in, bit-identical token out

- Tests inject the Gumbel noise tensor
- BF16 and FP32 logits, top-K bound 32 or 64, all softmax orders
- Production: Philox noise, fresh per launch



The cost: top-64 bound, one vocabulary size

- **Never samples outside the top 32/64**; top-k = 0 is not full-vocabulary
- **V is compiled in**: only 120,832 today

- Repetition penalty only: a bit cannot count tokens
- Temperature path: one stream per device
- Draft-token mask: temperature path only

Summary

- vLLM-style sampling: **~20** vocabulary-wide ops per step
- **Insight**: after top-k only ≤ 64 tokens matter; one read covers the rest
- Two kernels (one for temperature-only); penalty mask stays on GPU
- **4.6–8.5×** vs vLLM-style, **2.1–5.2×** vs FlashInfer

https://github.com/Tencent/hpc-ops